UNIVERSITI PUTRA MALAYSIA

M C L

Material Characterization Laboratory

PROVIDING QUALITY LAB SERVICES

Materials. Methods. Clear answers.

CAPABILITIES & SERVICES · 2026



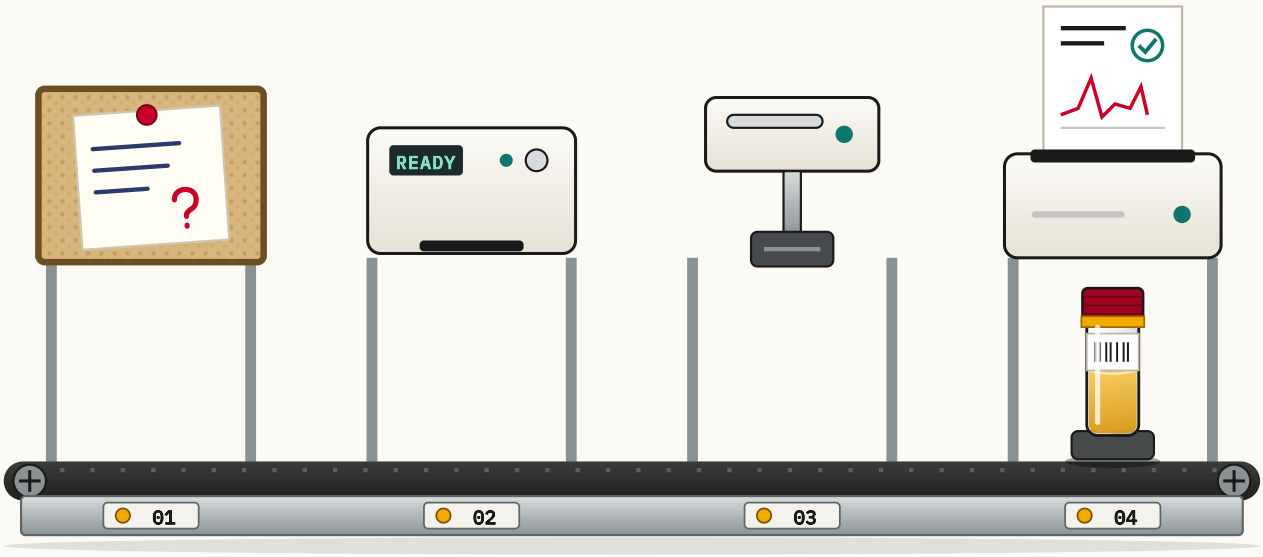
Choose an analytical method around the question that matters: composition, thermal behaviour, crystalline structure, surface morphology, particle size or separation of a mixture. MCL's service catalogue brings these forms of physical and chemical characterisation together for academic and industrial users.

This guide explains the published testing options, shows the listed prices and sets out how to begin a request. Discuss the material, the intended measurement and any reporting requirements with the laboratory before sending a sample. The final test selection and quotation require MCL's confirmation.

Material Characterization Laboratory
Department of Chemical and Environmental Engineering
Faculty of Engineering, Universiti Putra Malaysia
Serdang, Selangor.

WELCOME

Begin with the question your material must answer



01
Describe the question

02
Agree the method

03
Submit the sample

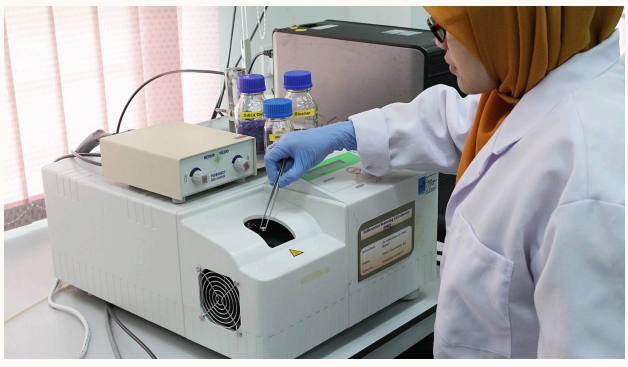
04
Receive the agreed output

Discuss requirements with MCL before submission.

MCL provides material characterisation and analytical testing within KKA at UPM's Faculty of Engineering. Its stated mission is to deliver reliable, high-quality services to academic and industrial partners, supported by expertise in analytical testing, metallurgy, polymers and advanced materials.

For a useful first enquiry, describe what the material is and what needs to be established. A question about melting behaviour leads to a different method from a question about elemental concentration or particle size. The published catalogue groups services by these measurement needs.

Send the sample description, requested analysis and reporting requirements to the quotation contacts. The current application form records applicant, payment and sample details; completion dates are agreed for the individual request.



KEY LINE Ask the measurement question first. Then select the test.

AT A GLANCE

A laboratory in numbers

MCL At a glance

Completed samples

32,000+

Reported ten-year track record

External clients

>300

Reported client network

Catalogue entries

19

technique / instrument entries

EKSA Best Laboratory

1st

Faculty of Engineering · 2024

Catalogue entries include detector configurations and combined methods.

MCL reports more than 32,000 completed samples over its ten-year track record and more than 300 external clients. These figures describe its reported service footprint across research and industry; they do not represent a count of accredited tests.

The published catalogue contains 19 technique and instrument entries, spanning thermal analysis, chromatography, spectroscopy, microscopy, elemental measurement, surface area, particle size and viscometry. Some entries are detector configurations or combined methods. The catalogue is therefore a guide to service choices rather than a claim of 19 separately accredited methods.

In June 2024, MCL received first place in the Faculty of Engineering EKSA Anugerah Makmal Terbaik. Assessment took place on 25–27 June, with the award presented on 28 June 2024.

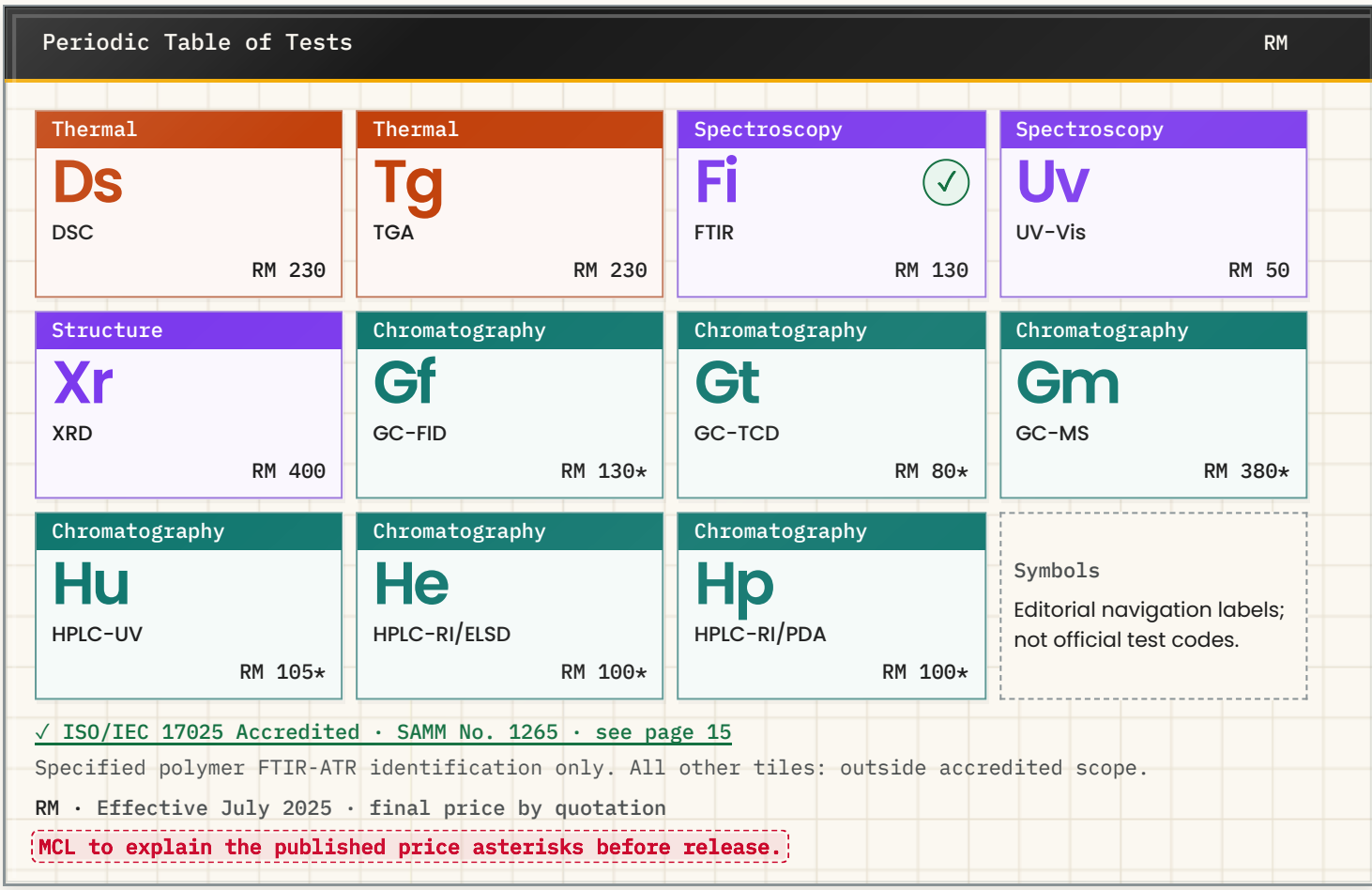


KEY FIGURE 32,000+ completed samples.

INDEX

PERIODIC TABLE OF TESTS

Find a test by the property



Read this table as a method index. The symbols are navigation labels created for this guide; they are not chemical element symbols or official laboratory test codes. Prices reproduce the published service list.

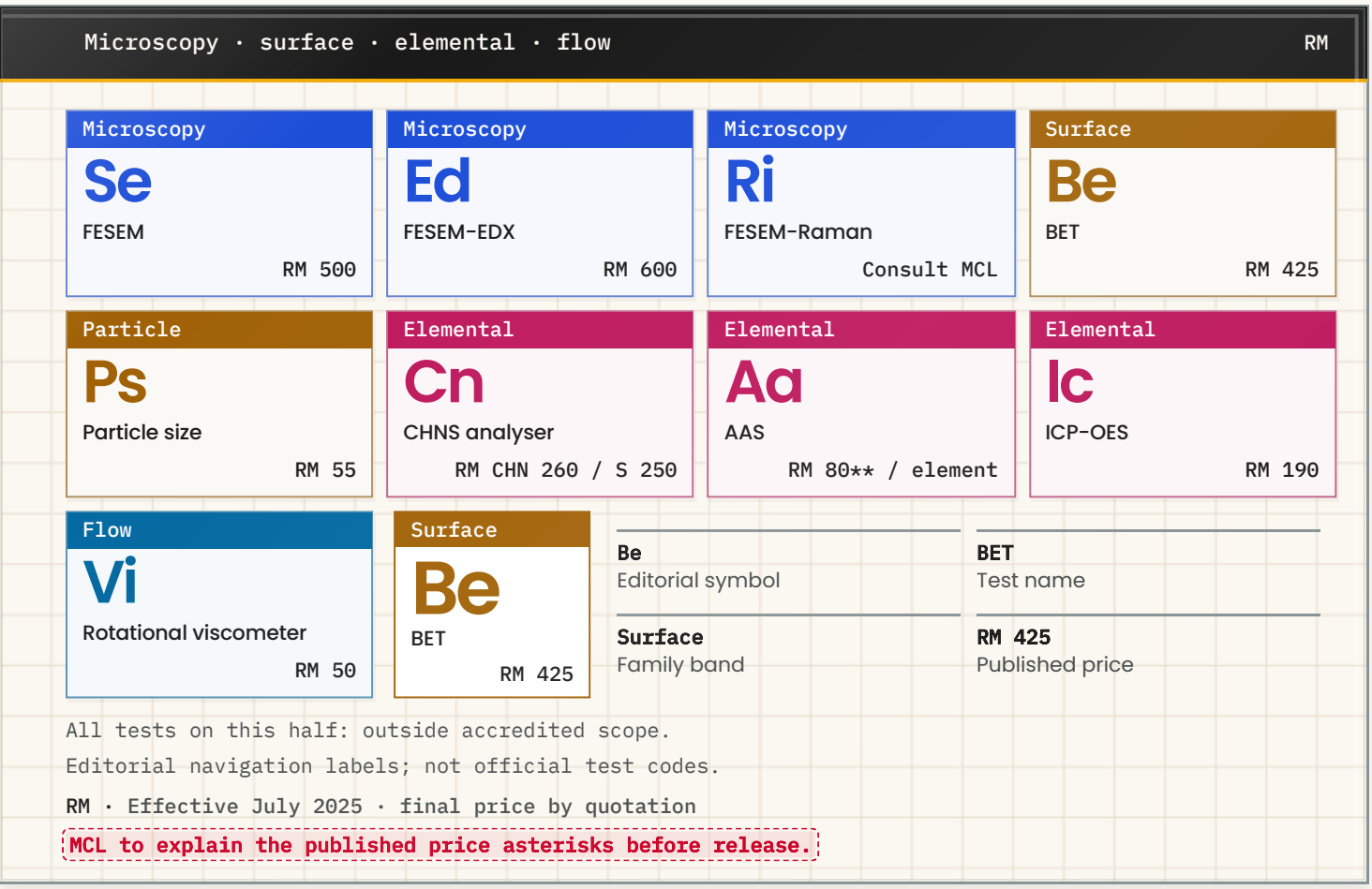
Begin with the property being investigated. Thermal methods follow heat flow or mass during heating. Spectroscopy examines interactions between matter and light. Chromatography separates components before a detector records them. These families guide the first conversation; the sample and required measurement determine the final selection.

The only accredited service represented here is qualitative FTIR-ATR identification of the specified polymers under SAMM 1265. Its full scope appears on page 15. All other tests are outside the accredited scope.

KEY LINE Effective July 2025 · final price by quotation.

PERIODIC TABLE OF TESTS

Connect the method to the material



Microscopy describes visible surface features at high magnification. Surface-area and particle-size measurements answer different questions: how much surface is accessible and how particle sizes are distributed. Elemental methods measure elemental content, while viscometry examines resistance to flow.

Combined services require careful selection. FESEM-EDX adds elemental information to electron microscopy. The FESEM-Raman option connects imaging with molecular information. The listed price for one configuration must not be treated as the price for a different detector, preparation or combined measurement.

Water analysis and other specialist options on the request form are introduced on page 14. They are not additional proof of equipment housed within MCL. Discuss the relevant laboratory, method, preparation and quotation before booking.

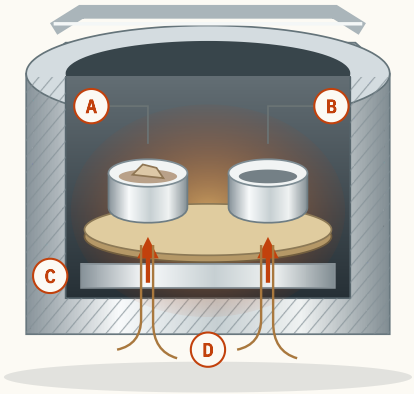
KEY LINE Different measurements can complement one another; they are not interchangeable.

THERMAL

Follow a material through temperature

DSC

RM 230

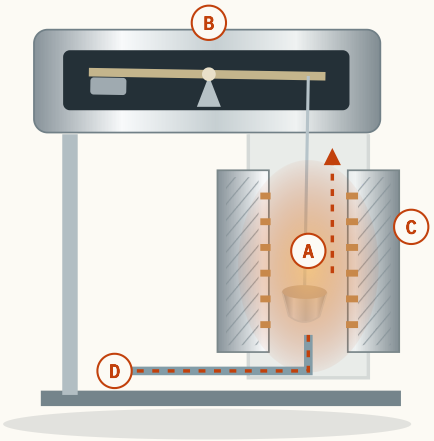


Heat absorbed / released

- **A** Sample pan
- **B** Reference pan
- **C** Controlled heating
- **D** Heat-flow measurement

TGA

RM 230



Mass change with temperature

- **A** Sample pan
- **B** Balance
- **C** Furnace
- **D** Mass measurement

Explanatory cutaways; no measured result is shown.

RM · Effective July 2025 · final price by quotation

Differential scanning calorimetry (DSC) · RM230. DSC records heat absorbed or released during a controlled temperature programme. MCL lists glass transition, melting, crystallisation, heat capacity and oxidation behaviour among its thermal-analysis applications.

Thermogravimetric analysis (TGA) · RM230. TGA records changes in sample mass as temperature changes. The listed applications include pyrolysis profiles and compositional analysis, helping users investigate thermal breakdown and material stability.

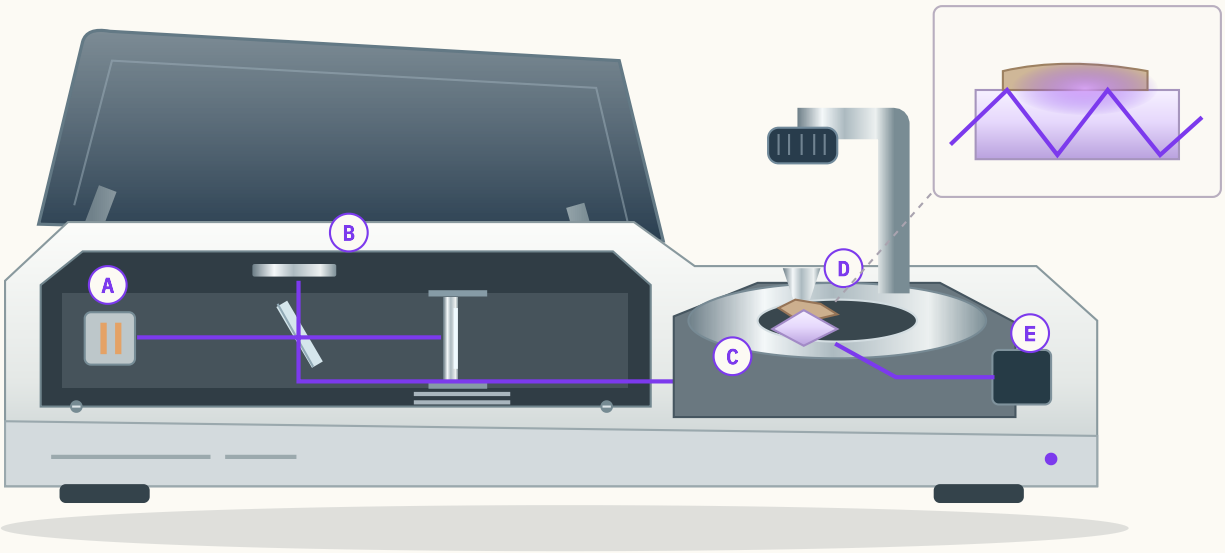
DSC and TGA provide complementary information: a thermal transition can occur without a corresponding mass change. Discuss which behaviour needs to be measured and the intended temperature programme when requesting analysis. Both services are outside the accredited scope.

KEY FIGURE DSC RM230 · TGA RM230. Effective July 2025 · final price by quotation.

SPECTROSCOPY · FTIR

Read the chemical fingerprint

Fourier-transform infrared spectroscopy (FTIR) · RM130. FTIR examines infrared absorption to help identify chemical groups. MCL lists ATR and KBr options; the sample and measurement question determine which preparation and technique are suitable.



A Infrared source B Interferometer C ATR crystal D Sample contact E Detector

Infrared field at the crystal surface

Explanatory schematic; no measured spectrum is shown.

RM 130 RM · Effective July 2025 · final price by quotation

In attenuated total reflectance (ATR), the sample interacts with an infrared field at the crystal surface. This page's cutaway explains that interaction; it is not a spectrum from a customer sample.

Accredited · SAMM 1265

specified qualitative polymer FTIR-ATR test only

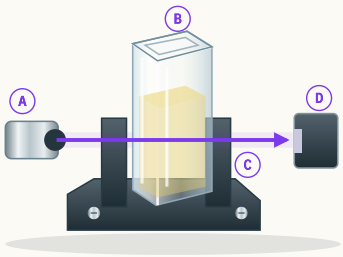
Accredited · SAMM 1265 applies only to the qualitative FTIR-ATR polymer test described in the exact scope below. Other FTIR services are outside the accredited scope.

> Engineering Testing Services, Faculty of Engineering, UPM (ETS FK) — Standards Malaysia SAMM No. 1265, MS ISO/IEC 17025:2017, chemical testing: qualitative FTIR-ATR identification of polymers (non-adhering solids and non-volatile liquids), 400–4000 cm⁻¹, method UPM/FK/ETS/TMD 1 based on ASTM E1252-98 (R2021). Accredited 7 Aug 2026 – 7 Aug 2031. Other tests are outside the accredited scope.

KEY FIGURE FTIR RM130. Effective July 2025 · final price by quotation.

SPECTROSCOPY · UV-VIS, XRD, RAMAN

Examine light, structure and chemistry

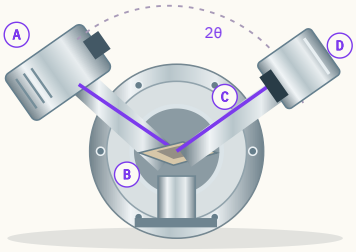


A Light source
B Cuvette
C Transmitted light
D Detector

UV-Vis

RM 50

UV-Vis spectrophotometry · RM50. UV-Vis measures the absorption of ultraviolet and visible light. Suitable applications include quantification of analytes that absorb in the relevant wavelength region.

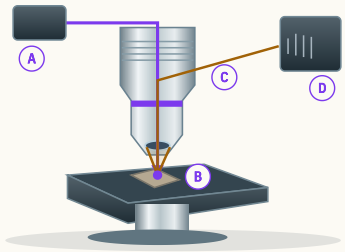


A X-ray source
B Crystalline sample
C Diffracted beam
D Detector

XRD

RM 400

Powder X-ray diffraction (XRD) · RM400. XRD examines diffraction from crystalline material to investigate structure and crystallinity. The published catalogue lists applications in solids and thin films.



A Laser
B Sample
C Scattered light
D Raman detector

FESEM-Raman

Consult MCL

Integrated FESEM-Raman imaging · consult MCL for price. Raman information can be combined with electron-microscope imaging to connect molecular characteristics with observed features. The catalogue mentions carbon materials, semiconductors, inorganic materials and minerals.

Explanatory schematics; no measured results are shown.

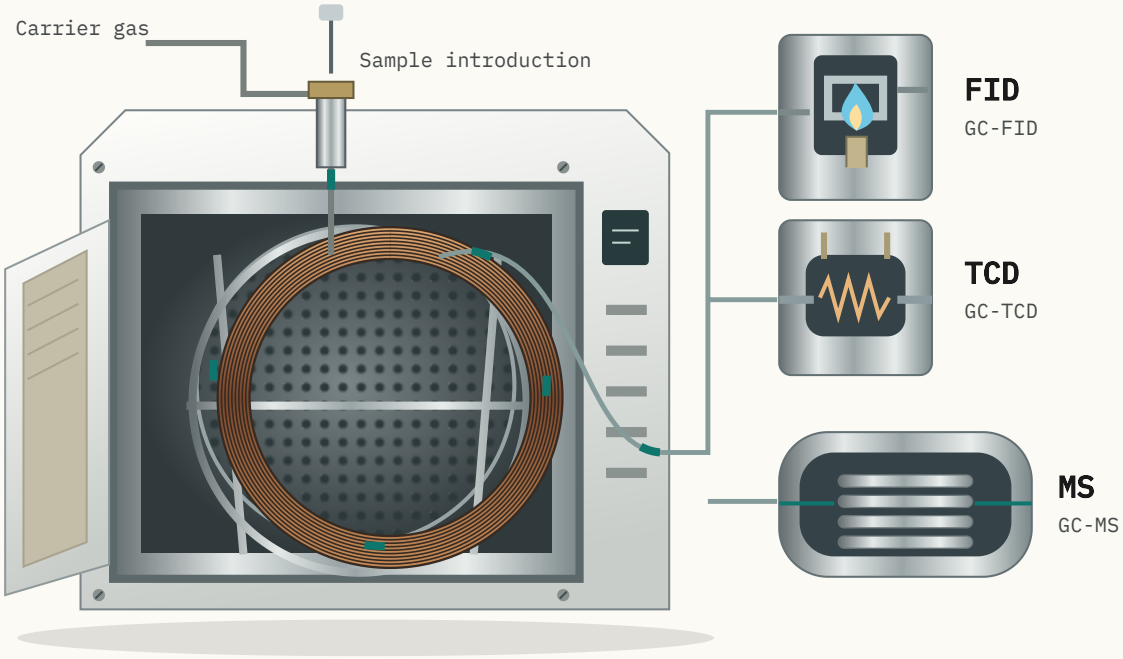
RM · Effective July 2025 · final price by quotation

These three options are outside the accredited scope. The illustration shows the measurement paths only; no sample spectrum, diffraction pattern or Raman result is presented.

KEY LINE Match absorption, crystalline structure or molecular information to the question.

CHROMATOGRAPHY · GC

Separate a complex mixture



Carrier gas · Sample introduction · Column separation · Detector

GC-FID
Flame-ionisation
RM130*

GC-TCD
Thermal-conductivity
RM80*

GC-MS
Mass spectrometer
RM380*

Separation schematic; coloured bands are explanatory labels, not experimental data.

RM · Effective July 2025 · final price by quotation

Gas chromatography with flame-ionisation detection (GC-FID) · RM130*. GC separates components before a flame detector records organic compounds. MCL lists an Agilent GC-FID service.

Gas chromatography–mass spectrometry (GC-MS) · RM380*. GC-MS separates volatile components and uses mass patterns to support identification. MCL lists a Shimadzu GC-MS service.

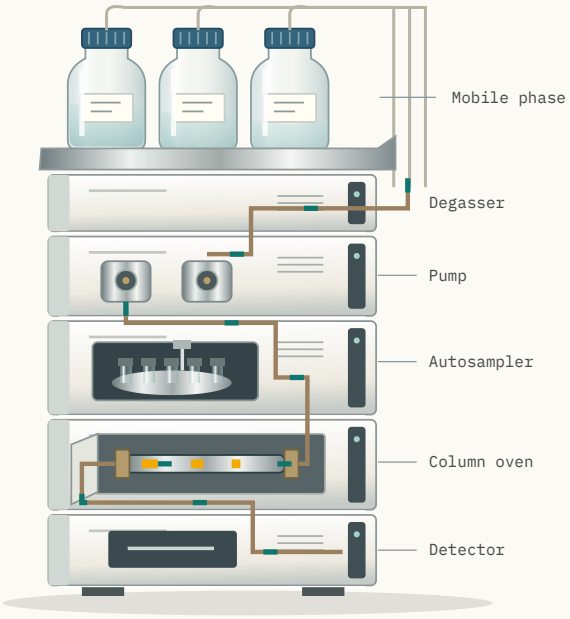
Gas chromatography with thermal-conductivity detection (GC-TCD) · RM80*. This configuration detects separated components through changes in thermal conductivity. MCL lists an Agilent GC-TCD service.

All three services are outside the accredited scope. Effective July 2025 · final price by quotation. The published asterisk is retained: **MCL to explain the GC price asterisk and quotation basis.**

KEY LINE One separation family; different detector information.

CHROMATOGRAPHY · HPLC

Choose detection for a liquid separation



Detector

HPLC-UV
Shimadzu
RM105*

HPLC-RI/ELSD
Agilent
RM100*

HPLC-RI/PDA
Jasco
RM100*

Mobile phase · Degasser · Pump · Autosampler / sample introduction · Column oven · Detector

Flow-path schematic; no measured separation is shown. Coloured bands are explanatory.

RM · Effective July 2025 · final price by quotation

High-performance liquid chromatography with UV detection (HPLC-UV) · RM105*. HPLC separates components carried in a liquid mobile phase. UV detection measures separated compounds that absorb ultraviolet light; the published service uses a Shimadzu system.

HPLC with refractive-index/photodiode-array detection (RI/PDA) · RM100*. RI and PDA offer different information from a liquid separation, including refractive-index response and UV/visible spectra. The published service uses a Jasco system.

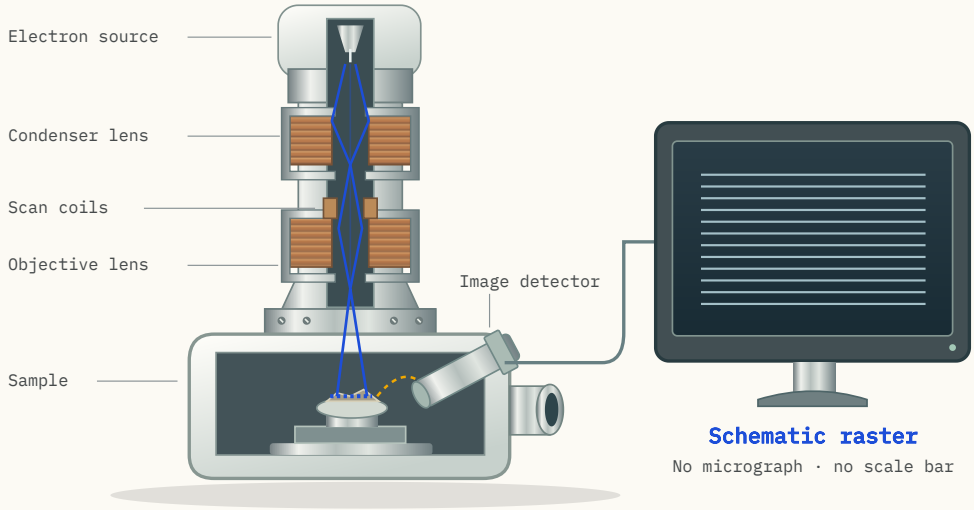
HPLC with refractive-index or evaporative light-scattering detection (RI/ELSD) · RM100*. These detector options record refractive-index changes or light scattered by particles formed after evaporation. MCL lists an Agilent HPLC RI/ELSD service.

All configurations are outside the accredited scope. Effective July 2025 · final price by quotation. **MCL to explain the HPLC price asterisk and which detector options the price covers.**

KEY LINE Describe the compounds and matrix before choosing the detector.

MICROSCOPY · FESEM

See surfaces and connect their chemistry



Electron source · Condenser lens · Scan coils · Objective lens
Focused beam · Raster scan · Sample · Image detector

FESEM	FESEM-EDX	FESEM-Raman
Surface image	+ EDX elemental information	+ Raman molecular information
RM500	**RM600**	**Consult MCL**

Explanatory instrument cutaway; no micrograph or scale claim is generated.

RM · Effective July 2025 · final price by quotation

Field-emission scanning electron microscopy (FESEM) · RM500. FESEM uses an electron beam to examine surface or cross-sectional features at high magnification. The published MCL catalogue identifies a Tescan system.

FESEM with integrated Raman laser imaging · consult MCL for price. This combined service connects electron-microscope images with molecular Raman information. Discuss the desired imaging and chemical information when requesting a quotation.

FESEM with energy-dispersive X-ray spectroscopy (FESEM-EDX) · RM600. EDX adds elemental information to the microscopy service. Choose it when the question concerns both observed features and their elemental composition.

These services are outside the accredited scope. Effective July 2025 · final price by quotation.

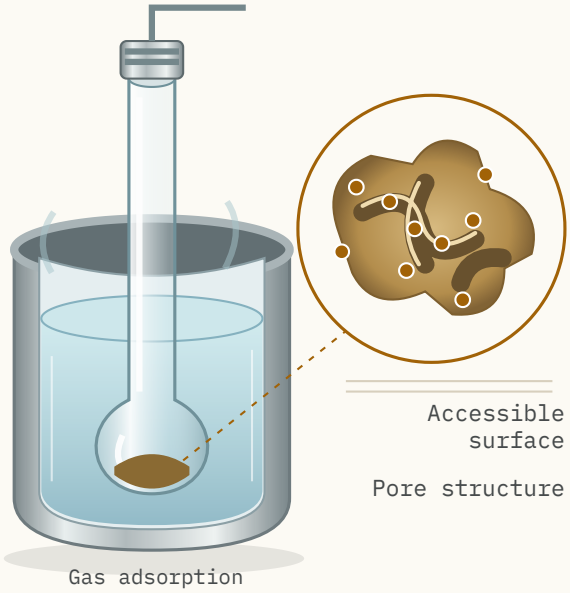
KEY FIGURE FESEM RM500 · FESEM-EDX RM600.

SURFACE & PARTICLE

Measure the spaces between particles

BET

RM425

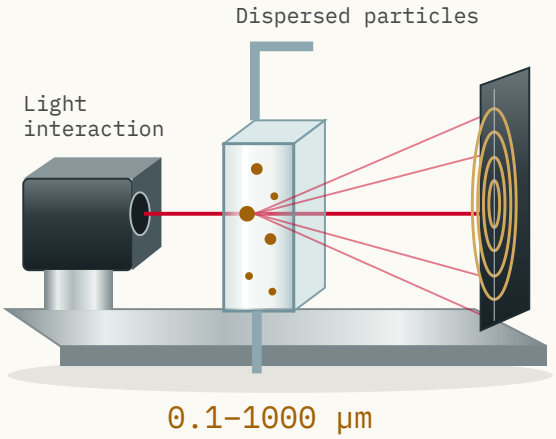


Gas adsorption

Accessible surface · Pore structure

Particle size

RM55



Light interaction

Dispersed particles · 0.1–1000 µm

Explanatory cutaways; no measured isotherm or distribution is shown.

Brunauer–Emmett–Teller surface-area analysis (BET) · RM425. Gas adsorption is used to investigate accessible surface area and pore structure. MCL lists a Micromeritics analyser; the application form also offers Langmuir analysis.

Particle-size analysis · RM55. The service measures the distribution of particle sizes rather than a single visual dimension. The published catalogue lists a Malvern analyser and a stated range of 0.1–1000 µm.

Surface area and particle size describe different aspects of a material. A particle-size distribution alone does not establish its pore structure, and surface area does not replace a size distribution. Discuss sample form and the information required before selecting one or both services.

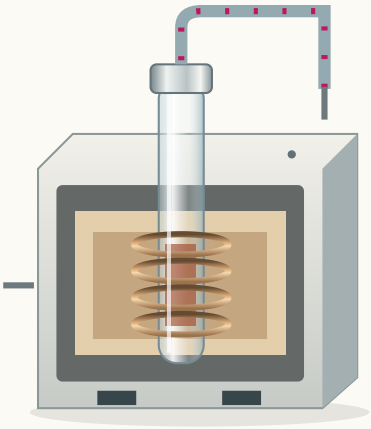
Both services are outside the accredited scope. Effective July 2025 · final price by quotation.

KEY FIGURE BET RM425 · particle-size analysis RM55.

ELEMENTAL

Ask which elements are present

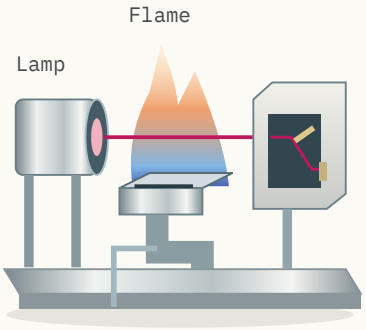
CHNS



CHN RM260 | S RM250

Combustion · Gas separation · Element measurement

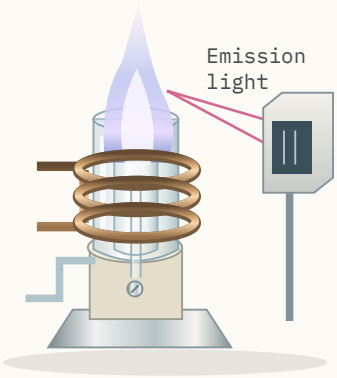
AAS



RM80** / element

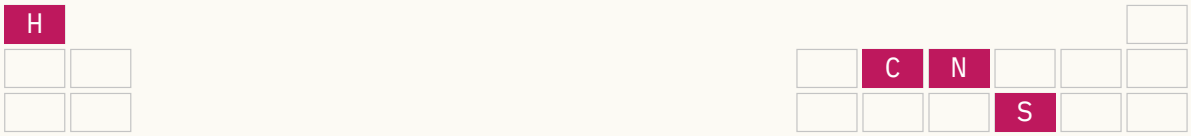
Lamp · Flame · Sample · Absorption detector

ICP-OES



RM190

Sample introduction · Plasma torch · Emission light · Detector



CHNS: carbon · hydrogen · nitrogen · sulphur

CHNS elemental analysis · CHN RM260; S RM250. The LECO analyser measures carbon, hydrogen, nitrogen and sulphur content. The published brochure lists CHN and sulphur as separate price entries.

Flame atomic absorption spectrophotometry (AAS) · RM80 per element.** AAS measures metal concentrations in liquid samples through element-specific light absorption. MCL lists a Thermo Scientific flame AAS service.

Inductively coupled plasma–optical emission spectroscopy (ICP-OES) · RM190. ICP-OES measures elemental concentrations from light emitted by a plasma. The published service uses a Perkin Elmer instrument.

These services are outside the accredited scope. Effective July 2025 · final price by quotation. **MCL to explain the AAS double asterisk and ICP-OES quotation basis.**

KEY LINE Specify the elements and sample matrix when asking for a quotation.

ELEMENT

WATER & OTHER

Match specialist requests to the right laboratory

REQUEST	ROUTE	PRICE
WATER ANALYSIS BOD · COD · TOC · Turbidity · Suspended solids · Dissolved solids · pH	Environmental Engineering Laboratory	Quotation
ROTATIONAL VISCOMETRY	Confirm with MCL	RM50
NIR/FIR	Confirm with MCL	Quotation
LIGHT MICROSCOPY	Confirm with MCL	Quotation
I-RAMAN	Confirm with MCL	Quotation
BOMB CALORIMETRY	Confirm with MCL	Quotation

Specialist requests — Outside accredited scope

Water analysis · request a quotation. The application form lists BOD, COD, TOC, turbidity, suspended/dissolved solids and pH. It assigns this group to the Environmental Engineering Laboratory; these request options are not a claim that the equipment is housed in MCL.

Rotational viscometry · RM50. A rotating spindle is used to examine a liquid's resistance to flow. This service appears in the published brochure.

NIR/FIR · request a quotation. Near- and far-infrared options appear on the application form. Discuss the wavelength region and required measurement before selecting either option.

Light microscopy · request a quotation. The form lists optical imaging. Confirm the imaging requirements and responsible laboratory before submission.

i-RAMAN · request a quotation. The form also lists this Raman option for molecular information. Confirm the measurement required and responsible laboratory before submission.

Bomb calorimetry · request a quotation. This method examines energy released by combustion. It appears as a request option, without a published price in the preferred brochure.

All options on this page are outside the accredited scope. **MCL to confirm availability, responsible laboratory and quotation basis for each unpriced option.**

KEY FIGURE Rotational viscometry RM50. Effective July 2025 · final price by quotation.

SPECIAL

ACCREDITATION

ISO/IEC 17025 Accredited

Engineering Testing Services, Faculty of Engineering, UPM (ETS FK) — Standards Malaysia SAMM No. 1265, MS ISO/IEC 17025:2017, chemical testing: qualitative FTIR-ATR identification of polymers (non-adhering solids and non-volatile liquids), 400–4000 cm^{-1}, method UPM/FK/ETS/TMD 1 based on ASTM E1252-98 (R2021). Accredited 7 Aug 2026 – 7 Aug 2031. Other tests are outside the accredited scope.

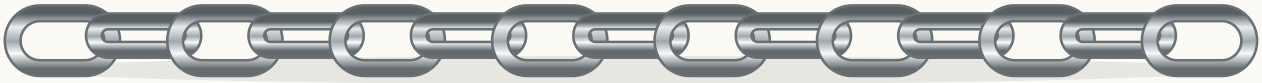
STANDARDS MALAYSIA ACCREDITED

SAMM No. 1265

MS ISO/IEC 17025:2017

Materials/Products Tested	Type of Test/Properties Measured/Range of Measurement	Standard Test Methods/Equipment/Techniques
Polymers Non-Adhering Solid and Non-Volatile Liquid	Infrared Spectra for Qualitative Analysis (Spectral range of 400 cm^{-1} to 4000 cm^{-1})	In-house Method UPM/FK/ETS/TMD 1 based on ASTM E1252-98 (Reapproved 2021) using FTIR-ATR

Editorial scope presentation; not a reproduction of a certificate.



SI · National standards · Standards Malaysia · ETS FK / MCL · Your report

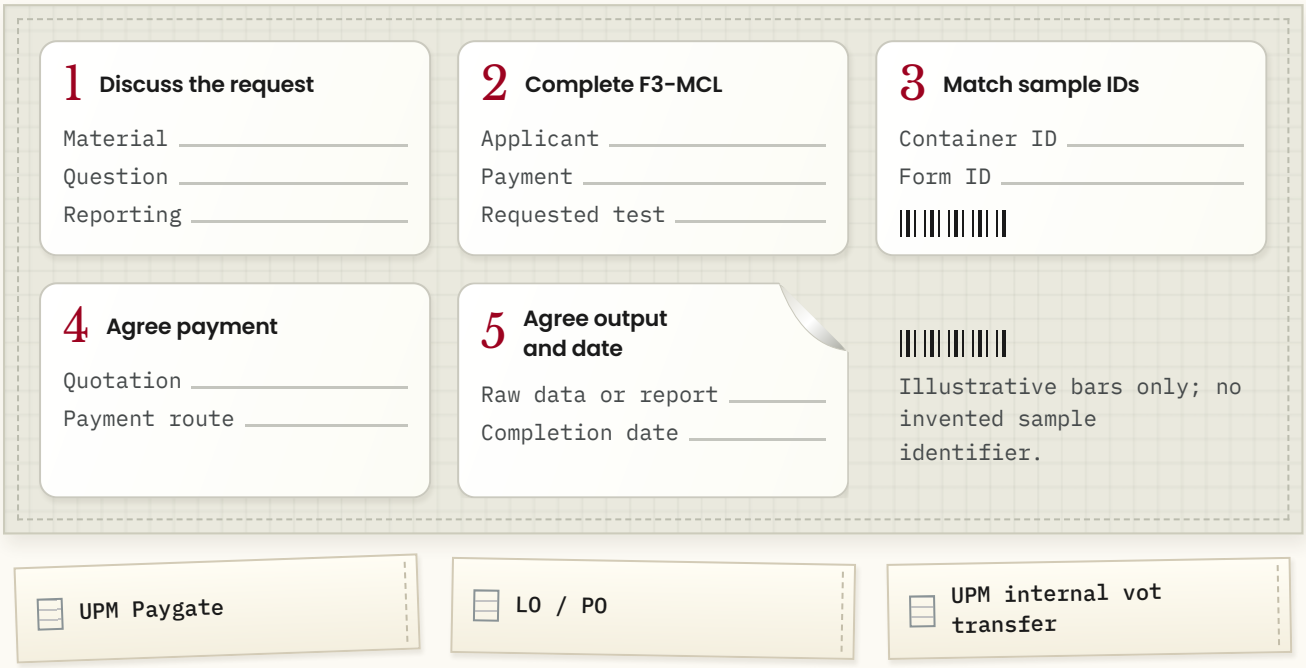
Conceptual traceability chain; not an individual calibration record.

Check the current Standards Malaysia schedule and state any accredited-report requirement when making the request. The current application form includes an accredited-testing review; a test report is compulsory for MS ISO/IEC 17025 testing.

KEY FIGURE SAMM 1265 · FTIR-ATR polymers only.

HOW TO SEND A SAMPLE

Prepare a sample that can be accepted



1. Discuss the request. Email the sample description, intended analysis and reporting needs. MCL's named quotation contacts are Dr. Dayang Radiah Awang Biak and ChM. Shafizah Masuri.

2. Complete the current form. Use F3-MCL, effective 18 June 2026, issue 1, revision 3. Record the applicant, payment details, samples and requested services.

3. Identify every sample. Use the form's sample-ID sheet and keep each sample identifier consistent across the containers and request. Confirm preparation, quantity and delivery arrangements with MCL. **MCL to confirm technique-specific quantities, containers and acceptance instructions.**

4. Agree payment arrangements. Published routes include the UPM payment gateway, LO/PO and UPM internal vot transfer. Follow the confirmed quotation and payment instructions.

5. Agree the output and completion date. Select raw data or a test report as applicable; an accredited test requires a report. Completion is agreed individually.

KEY LINE Confirm acceptance before delivery.

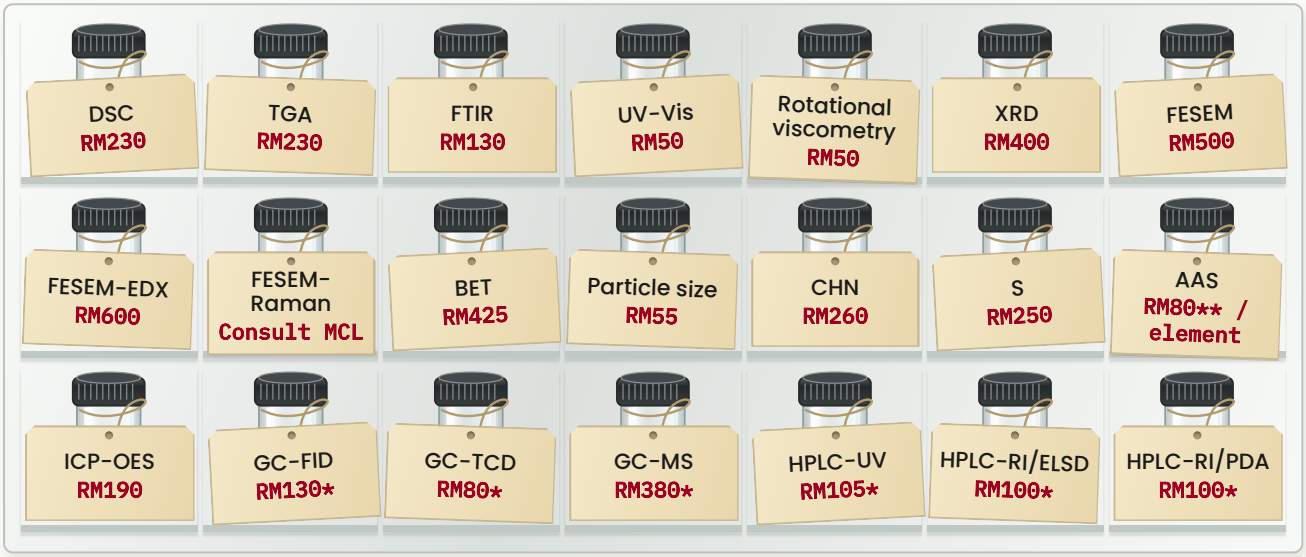
SERVICE

PRICES

Plan the test, then request a quotation

Use the published prices to prepare an enquiry. Confirm the sample preparation, detector configuration, measurement and required output before accepting a quotation. A listed instrument price does not establish that every application or additional preparation is included.

Effective July 2025 · final price by quotation.



Service	Published price (RM)
DSC / TGA	230 each
FTIR	130
UV-Vis / rotational viscometry	50 each
XRD	400
FESEM / FESEM-EDX	500 / 600
FESEM-Raman	Consult MCL
BET / particle-size analysis	425 / 55
CHN / sulphur	260 / 250
AAS	80** per element
ICP-OES	190
GC-FID / GC-TCD / GC-MS	130* / 80* / 380*
HPLC-UV / HPLC-RI/ELSD / HPLC-RI/PDA	105* / 100* / 100*

The published single and double asterisks are retained. **MCL to explain all asterisks, price units and preparation inclusions before release.** Water and other unpriced request options require a quotation.

The prices include services outside the accredited scope. Only the specified polymer FTIR-ATR test is accredited under SAMM 1265; see page 15.

KEY LINE Send the test list with the sample description to kkamcl1595@upm.edu.my.

TRAINING

Learn the method at the instrument

April 2027

13–14

DSC

Two-day hands-on workshop
Faculty of Engineering, UPM

Early bird until 31 December 2026	RM700
Normal Two-day workshop	RM800
One day	RM400

Registration closes **31 March 2027**



DSC · 13–14 April 2027



FTIR · Historical announcement



FESEM · Historical announcement

Thermal Analysis Using Differential Scanning Calorimetry (DSC) is advertised as a two-day hands-on workshop on **13–14 April 2027**, at the Faculty of Engineering, UPM. Published fees are **RM700 early bird until 31 December 2026**, **RM800 normal**, or **RM400 for one day**. Registration closes **31 March 2027**.

The public workshop record also includes “Discover the Invisible: FTIR Spectroscopy Workshop”, advertised for 1–2 April 2026, and “Dive into the Nanoscale: Demo FESEM Workshop”, advertised for 22 July 2025. The FESEM programme listed live demonstrations, sample preparation and data interpretation. These are historical announcements, not current booking offers.

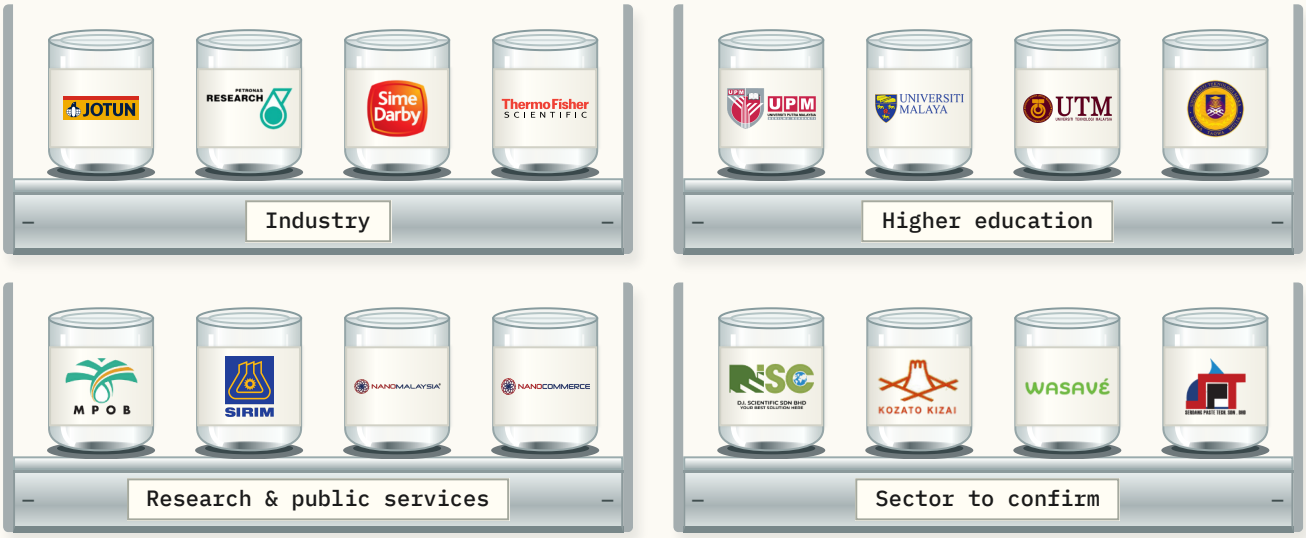
Ask MCL about the DSC programme, remaining places, registration route and suitability for the intended learning needs. Contact **kkamcl1595@upm.edu.my** or **+60 3 9769 1595**. **MCL to confirm the current DSC registration link and availability before publication.**

SERVICE

KEY FIGURE DSC hands-on workshop · 13–14 April 2027.

CLIENTS + FEEDBACK

A network across research and industry



16 of 80 clients on MCL's client sheet are shown. No sector counts are inferred.
Confirm the sector of the last rack.

MCL reports more than 300 external clients. Its supplied client wall includes organisations across industry, higher education, research and public services. The wall records client identities; it does not imply endorsement of this publication or accreditation of every service used. **MCL to confirm permission to publish every supplied client logo and identify ambiguous marks.**

4.5 ★ · 16 reviews (Oct 2026). View the Google listing.

"UPM laboratory fully equipped"
— Qu Shi, Google review, 2019.

"Breakthrough Science with Academic Excellence"
— M.Y.P Research in Motion, Google review, 2018.

"Saya tidak pernah hantar, terasa lah nak hantar sampel, dengar macam best"
"I have never sent a sample; I feel like sending one — it sounds good."
— S. Noor, UPM staff; English translation. This is an expression of interest, not a completed-service review. **Confirm S. Noor's consent to publish the quote and translation.**

Share feedback by email while the feedback form is prepared. **MCL to provide the feedback form URL and publication-consent wording.**

KEY FIGURE More than 300 external clients.

SUSTAINABILITY

Good science includes responsible practice



Bottle shapes organise the themes; fill levels do not represent measured performance.

Environmental care, people and accountable practice all matter in an analytical laboratory. Responsible operation includes careful handling of chemicals, appropriate waste segregation and sensible use of energy. These principles explain the E, S and G labels shown here; they are not measured claims about MCL's environmental performance.

The public record includes a KKA STEM outreach programme at SMK Convent Kajang, documented in an official gallery published on 6 December 2024. The event date is not stated. This is departmental outreach evidence, rather than proof that every pictured participant represents MCL.

MCL's published workshop announcements provide a further route into analytical learning. Its public brochure makes service prices available, while the current SAMM 1265 schedule defines the accredited polymer FTIR-ATR test separately from the wider catalogue.

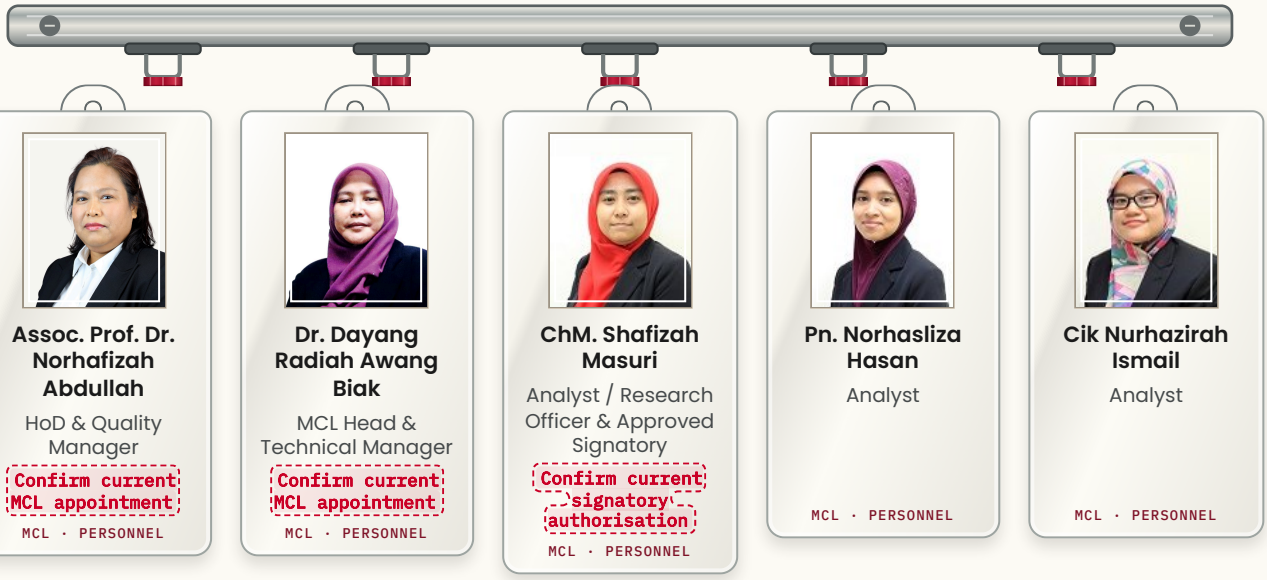
MCL to confirm implemented environmental practices and provide approved evidence before any quantified environmental claim is added.

ABOUT

KEY LINE Show the practice. State the evidence. Keep the scope clear.

TEAM

Meet the people behind the service



Assoc. Prof. Dr. Norhafizah Abdullah
HoD & Quality Manager
Confirm current MCL appointment
MCL · PERSONNEL

Dr. Dayang Radiah Awang Biak
MCL Head & Technical Manager
Confirm current MCL appointment
MCL · PERSONNEL

ChM. Shafizah Masuri
Analyst / Research Officer & Approved Signatory
Confirm current signatory authorisation
MCL · PERSONNEL

Pn. Norhasliza Hasan
Analyst
MCL · PERSONNEL

Cik Nurhazirah Ismail
Analyst
MCL · PERSONNEL

ASSOCIATE MEMBERS — Listed roles and assignments to confirm where shown



Dr. Mohd Yusof Harun
Deputy Technical Manager



En. Norisham Abdul Wahab
Analyst
Confirm MCL assignment.



Pn. Fatin Nordalila Omar
Document Control Officer
Confirm MCL assignment.



Pn. Hartini Mansor
Assistant Engineer



En. Adli Nazri Mohd Kassim
Confirm MCL role and assignment.

Portraits: Universiti Putra Malaysia, Faculty of Engineering staff pages.

MCL lists **Assoc. Prof. Dr. Norhafizah Abdullah** as Head of Department and Quality Manager, and **Dr. Dayang Radiah Awang Biak** as Head of MCL and Technical Manager. The public department record confirms Norhafizah as HoD and identifies Dayang Radiah as a quotation contact. **MCL to confirm the current laboratory appointment list.**

The listed analytical team comprises **ChM. Shafizah Masuri**, Analyst/Research Officer and Approved Signatory; **Pn. Norhasliza Hasan**, Analyst; and **Cik Nurhazirah Ismail**, Analyst. The public staff record locates all three at MCL and lists Shafizah as Research Officer.

Associate members are **Dr. Mohd Yusof Harun**, Deputy Technical Manager; **En. Norisham Abdul Wahab**, Analyst; **Pn. Fatin Nordalila Omar**, Document Control Officer; **Pn. Hartini Mansor**, Assistant Engineer; and **En. Adli Nazri Mohd Kassim**, whose MCL role is **confirm**. **Confirm Norisham, Fatin and Adli Nazri's MCL assignments, which differ from public staff-page assignments; confirm current associate appointments.**

KEY LINE — For quotations, contact Dr. Dayang Radiah or ChM. Shafizah Masuri.

TEAM & MASCOT

Remember the people. Meet the water monitor.

MCL · LABORATORY REGISTER 01 / 02

Name	Role	Period
En. Amir Syarifuddeen	confirm	confirm
En. Mustapha Akil	confirm	confirm
Dr. Magaret Sivapragasam	confirm	confirm
Pn. Rohana	confirm	confirm

PERSONNEL · ARCHIVE FOLIO 21

MCL · LABORATORY REGISTER 02 / 02

Name	Role	Period
En. Mohd Hashim Saad	confirm	confirm
En. Ismail Abdul Ghani	confirm	confirm
Cik Syazwani Abdul Halim	confirm	confirm
Pn. Nor Hasni Zahari	confirm	confirm

PERSONNEL · ARCHIVE FOLIO 22



WATER MONITOR
[Mascot name — MCL to choose]

MCL recognises the contribution of past members to its laboratory community. Its supplied list names **En. Amir Syarifuddeen, En. Mustapha Akil, Dr. Magaret Sivapragasam, Pn. Rohana, En. Mohd Hashim Saad, En. Ismail Abdul Ghani, Cik Syazwani Abdul Halim and Pn. Nor Hasni Zahari**. **MCL to confirm all past-member names, spellings, roles and service periods; the supplied spelling "Magaret" is retained pending confirmation.**

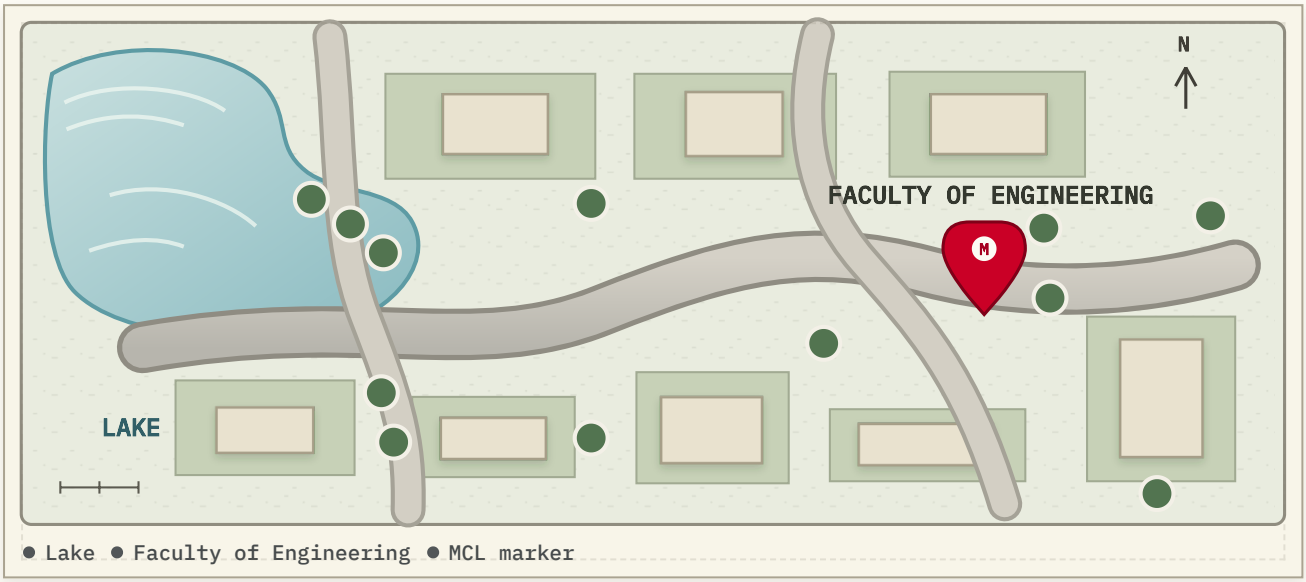
The laboratory's mascot is a **water monitor**, inspired by the animals around the Faculty of Engineering lake. Its white laboratory coat, magnifier, glasses and yellow MCL badge connect a familiar campus animal with close observation and practical science.

The character offers a welcoming guide through methods and sample preparation. Its name remains **[Mascot name — MCL to choose]**. **MCL to choose the mascot's name.**

KEY LINE Curiosity, observation and a connection to campus life.

CONTACT

Speak to MCL before sending a sample



DESTINATION MCL · Block J2.1



ILLUSTRATIVE DOOR SIGN

MATERIAL CHARACTERIZATION LABORATORY
MCL · Block J2.1

VISIT THE LABORATORY — HOURS

Monday–Friday 08:00–17:00

Lunch · Mon–Thu	13:00–14:00
Lunch · Friday	12:15–14:45

Published department hours; confirm delivery arrangements.

Illustrative campus map; use linked Google Maps for navigation.

Material Characterization Laboratory
Block J2.1 **MCL to confirm block notation J2.1 versus J.21**
Department of Chemical and Environmental Engineering
Faculty of Engineering, Universiti Putra Malaysia
43400 UPM Serdang, Selangor, Malaysia.

General enquiries: kkamcl1595@upm.edu.my · **+60 3 9769 1595**.
Quotations: Dr. Dayang Radiah Awang Biak, dradiah@upm.edu.my; ChM. Shafizah Masuri, shafizah_m@upm.edu.my.

Published department hours are **Monday–Friday, 08:00–17:00**. Lunch is **13:00–14:00 Monday–Thursday** and **12:15–14:45 Friday**. Confirm sample delivery and instrument arrangements before travelling. The published hours are not a self-service booking schedule.

Open the laboratory's Google Maps listing. **MCL to confirm the preferred gate, parking, delivery point and accessible route for the illustrated campus map.**

KEY LINE Email the material, measurement question and sample count with the enquiry.

Your next material question starts here

Bring the question, the material description and the intended result. MCL's published services cover physical and chemical characterisation, with quotation contacts available through its KKA laboratory page. Use email to discuss the method, sample requirements and reporting needs before submission.

01 EXPLORE MCL

QR code after the link is verified

mcl-upm.github.io

Verify published website before QR generation.

Explore the services.

Open the website to browse the service catalogue and contact details.

02 SHARE FEEDBACK

QR code after the link is verified

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MCL to supply approved feedback form URL.

Email feedback for now

Share your feedback.

Help the laboratory understand the experience of making an enquiry, submitting samples and receiving the agreed output. The feedback route requires MCL's approved form.

03 OPEN THIS GUIDE

QR code after the link is verified

–

Insert the published public customer flipbook URL.

Keep this guide.

Save the public flipbook for the next discussion about testing. The three QR labels below must link to the approved website, feedback form and this customer edition. Verify all three live destinations before generating QR codes.



KEY CONTACT

kkamcl1595@upm.edu.my · +60 3 9769 1595.

MATERIAL CHARACTERIZATION LABORATORY · FACULTY OF ENGINEERING · UPM SERDANG